

# Technical Compilation and Calculation Annex

Consolidated CBAM evidence dossier and preparation statements

Report report\_aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa · Release 5

AUTOMATED PREPARATION CHECKS  
PASSED

CONFIDENTIAL - VERIFIER PREPARATION WORKSPACE

**DOCUMENT BOUNDARY** This package supports preparation for independent verification. It is not a verification opinion, accreditation decision, customs decision, CBAM Registry submission or acceptance guarantee.

| Control field |                                                                  |                 |                          |
|---------------|------------------------------------------------------------------|-----------------|--------------------------|
| Case ID       | case_verifier_grade_fixture                                      | Generated       | 2026-07-16T12:00:00.000Z |
| Ruleset       | 3.0.0                                                            | Source registry | CBAM-EU-2026.07.16       |
| Root hash     | 7e12ba9592177cc10e0e4541bb103905e0345303fa794949881b86ec2ab48f61 | Verifier status | NOT_REVIEWED             |

## Product Scope - Controlled identity

| Field               |                                        |
|---------------------|----------------------------------------|
| Importer            | CBAMValid Importer B.V.                |
| EORI                | NL123456789AB                          |
| Exporter / operator | Verified Steel Operator GmbH           |
| Installation        | Verified Integrated Steel Installation |
| Country             | DE                                     |
| Production route    | Blast Furnace Route (BF-BOF)           |
| Reporting period    | 2026-ANNUAL                            |

## Product Scope - Goods population

| Good |          |                |    |     |    |     |      |
|------|----------|----------------|----|-----|----|-----|------|
| 1    | 72011011 | IRON_AND_STEEL | 60 | 0.6 | 72 | 1.2 | 0.06 |
| 2    | 72011019 | IRON_AND_STEEL | 40 | 0.4 | 48 | 1.2 | 0.06 |

## Product Scope - Scope conclusion

**AUTOMATED READINESS** READY\_FOR\_INDEPENDENT\_VERIFICATION

## Product Scope - Legal basis

| Source        |            |            |                                                                                                                                                                                                  |
|---------------|------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REG_2023_956  | 32023R0956 | 2023-05-17 | CBAM framework; Annex I goods scope; Annex IV embedded-emissions calculation framework                                                                                                           |
| REG_2025_2083 | 32025R2083 | 2025-10-20 | 50 tonne annual mass threshold for cement, fertilisers, iron and steel, and aluminium; definitive-period declaration and certificate obligations; actual values require independent verification |

# Technical Compilation and Calculation Annex

Consolidated CBAM evidence dossier and preparation statements

Report report\_aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa · Release 5

AUTOMATED PREPARATION CHECKS  
PASSED

CONFIDENTIAL - VERIFIER PREPARATION WORKSPACE

| Source         |            |            |                                                                                                                                               |
|----------------|------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| IMPL_2025_2546 | 32025R2546 | 2026-01-01 | reasonable assurance and risk-based verification; 5 percent per-good materiality levels; electronic verification report template              |
| IMPL_2025_2547 | 32025R2547 | 2026-01-01 | functional units and production processes; monitoring plan minimum elements; sector-specific system boundaries and embedded-emissions methods |
| IMPL_2025_2548 | 32025R2548 | 2026-01-01 | calculation and publication of CBAM certificate prices                                                                                        |
| DEL_2025_2551  | 32025R2551 | 2026-01-01 | accreditation and competence of verifiers; verification planning, evidence and site-visit framework; oversight and independence               |

## CN Code - Classification register

| Good |          |                |                                          |                                          |
|------|----------|----------------|------------------------------------------|------------------------------------------|
| 1    | 72011011 | IRON_AND_STEEL | 11111111-1111-4111-8111-11111111<br>1111 | 11111111-1111-4111-8111-11111111<br>1111 |
| 2    | 72011019 | IRON_AND_STEEL | 11111111-1111-4111-8111-11111111<br>1111 | 11111111-1111-4111-8111-11111111<br>1111 |

## CN Code - Classification boundary

This document records the classification supplied in the sealed case. It does not replace a binding customs classification decision. Any disputed or uncertain CN code remains subject to customs review.

## Checklist - Monitoring-plan requirements

| Requirement |                                                                 |            |                                                                                                                                                    |
|-------------|-----------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| MP-01       | Monitoring-plan date, version and responsible operator identity | DOCUMENTED | Case version 1; owner recorded                                                                                                                     |
| MP-02       | Installation description and production processes               | DOCUMENTED | Blast Furnace Route (BF-BOF)                                                                                                                       |
| MP-03       | CN-coded goods and functional production units                  | DOCUMENTED | 2 good(s) recorded                                                                                                                                 |
| MP-04       | System boundaries and directly connected processes              | DOCUMENTED | Coke preparation, sinter plant, blast furnace, basic oxygen furnace, casting and finishing operations within the controlled installation boundary. |
| MP-05       | Source streams, emission sources and measurement methods        | DOCUMENTED | 0 precursor(s); direct and electricity inputs recorded                                                                                             |
| MP-06       | Data-source hierarchy, corroboration and control activities     | DOCUMENTED | 1 approved and clean evidence file(s)                                                                                                              |
| MP-07       | Calculation methods, conversions and rounding                   | DOCUMENTED | 8 hashed trace node(s)                                                                                                                             |
| MP-08       | Goods allocation method and reconciliation                      | DOCUMENTED | Allocation delta 0                                                                                                                                 |
| MP-09       | Precursor scope and embedded emissions treatment                | DOCUMENTED | Accepted no-precursor decision                                                                                                                     |
| MP-10       | Methodology deviations, estimates and technical justifications  | DOCUMENTED | 2/2 methodology decision(s) accepted                                                                                                               |
| MP-11       | Per-good materiality assessment                                 | DOCUMENTED | 5% per-good specific-emissions threshold                                                                                                           |
| MP-12       | Evidence integrity and immutable calculation provenance         | DOCUMENTED | Calculation root 7e12ba9592177cc10e0e4541bb103905e0345303fa794949881b86ec2ab48f61                                                                  |

# Technical Compilation and Calculation Annex

Consolidated CBAM evidence dossier and preparation statements

Report report\_aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa · Release 5

AUTOMATED PREPARATION CHECKS  
PASSED

CONFIDENTIAL - VERIFIER PREPARATION WORKSPACE

## Checklist - Automated control findings

| Rule    |                                              |                |                      |                             |
|---------|----------------------------------------------|----------------|----------------------|-----------------------------|
| QC_00   | Operator, installation and boundary identity | PASS           | No automated finding | REM_COMPLETE_CASE_IDENTITY  |
| QC_01   | EORI format and evidence                     | PASS           | No automated finding | —                           |
| QC_02   | Definitive-period reporting year             | PASS           | No automated finding | REM_CORRECT_REPORTING_YEAR  |
| QC_03_0 | Good 1 CN code and evidence                  | PASS           | No automated finding | —                           |
| QC_04_0 | Good 1 production and evidence               | PASS           | No automated finding | —                           |
| QC_05_0 | Good 1 sector                                | PASS           | No automated finding | REM_SELECT_SUPPORTED_SECTOR |
| QC_03_1 | Good 2 CN code and evidence                  | PASS           | No automated finding | —                           |
| QC_04_1 | Good 2 production and evidence               | PASS           | No automated finding | —                           |
| QC_05_1 | Good 2 sector                                | PASS           | No automated finding | REM_SELECT_SUPPORTED_SECTOR |
| QC_05A  | Allocation, evidence and reconciliation      | PASS           | No automated finding | —                           |
| QC_06   | directEmissions value, unit and evidence     | PASS           | No automated finding | —                           |
| QC_07   | electricityConsumed value, unit and evidence | PASS           | No automated finding | —                           |
| QC_08   | gridEmissionFactor value, unit and evidence  | PASS           | No automated finding | —                           |
| QC_09   | Precursor scope                              | PASS           | No automated finding | REM_CONFIRM_PRECURSOR_SCOPE |
| QC_10   | Evidence integrity                           | PASS           | No automated finding | —                           |
| QC_11   | Carbon price records                         | NOT_APPLICABLE | No automated finding | —                           |

## Checklist - Evidence coverage

| Measure                          |      |
|----------------------------------|------|
| Registered evidence files        | 1    |
| Approved and malware-clean files | 1    |
| Linked input fields              | 18   |
| Linked calculation nodes         | 5    |
| Coverage rate                    | 100% |
| Duplicate hashes                 | 0    |

# Technical Compilation and Calculation Annex

Consolidated CBAM evidence dossier and preparation statements

Report report\_aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa · Release 5

AUTOMATED PREPARATION CHECKS  
PASSED

CONFIDENTIAL - VERIFIER PREPARATION WORKSPACE

## Monitoring Plan - Installation and boundary

Coke preparation, sinter plant, blast furnace, basic oxygen furnace, casting and finishing operations within the controlled installation boundary.

Production route: Blast Furnace Route (BF-BOF)

## Monitoring Plan - Minimum monitoring-plan elements

| Requirement |                                                                 |            |                                                                                                                                                    |
|-------------|-----------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| MP-01       | Monitoring-plan date, version and responsible operator identity | DOCUMENTED | Case version 1; owner recorded                                                                                                                     |
| MP-02       | Installation description and production processes               | DOCUMENTED | Blast Furnace Route (BF-BOF)                                                                                                                       |
| MP-03       | CN-coded goods and functional production units                  | DOCUMENTED | 2 good(s) recorded                                                                                                                                 |
| MP-04       | System boundaries and directly connected processes              | DOCUMENTED | Coke preparation, sinter plant, blast furnace, basic oxygen furnace, casting and finishing operations within the controlled installation boundary. |
| MP-05       | Source streams, emission sources and measurement methods        | DOCUMENTED | 0 precursor(s); direct and electricity inputs recorded                                                                                             |
| MP-06       | Data-source hierarchy, corroboration and control activities     | DOCUMENTED | 1 approved and clean evidence file(s)                                                                                                              |
| MP-07       | Calculation methods, conversions and rounding                   | DOCUMENTED | 8 hashed trace node(s)                                                                                                                             |
| MP-08       | Goods allocation method and reconciliation                      | DOCUMENTED | Allocation delta 0                                                                                                                                 |
| MP-09       | Precursor scope and embedded emissions treatment                | DOCUMENTED | Accepted no-precursor decision                                                                                                                     |
| MP-10       | Methodology deviations, estimates and technical justifications  | DOCUMENTED | 2/2 methodology decision(s) accepted                                                                                                               |
| MP-11       | Per-good materiality assessment                                 | DOCUMENTED | 5% per-good specific-emissions threshold                                                                                                           |
| MP-12       | Evidence integrity and immutable calculation provenance         | DOCUMENTED | Calculation root 7e12ba9592177cc10e0e4541bb103905e0345303fa794949881b86ec2ab48f61                                                                  |

## Monitoring Plan - Input hierarchy

| Input                |     |           |         |                                    |                                                                |
|----------------------|-----|-----------|---------|------------------------------------|----------------------------------------------------------------|
| Direct emissions     | 80  | tCO2e     | PRIMARY | 11111111-1111-4111-8111-1111111111 | Documented direct measurement and reconciled production ledger |
| Electricity          | 100 | MWh       | PRIMARY | 11111111-1111-4111-8111-1111111111 | Documented direct measurement and reconciled production ledger |
| Grid emission factor | 0.4 | tCO2e/MWh | PRIMARY | 11111111-1111-4111-8111-1111111111 | Documented direct measurement and reconciled production ledger |

# Technical Compilation and Calculation Annex

Consolidated CBAM evidence dossier and preparation statements

Report report\_aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa · Release 5

AUTOMATED PREPARATION CHECKS  
PASSED

CONFIDENTIAL - VERIFIER PREPARATION WORKSPACE

## Process Map - End-to-end process

| Stage |                                                                              |                                            |
|-------|------------------------------------------------------------------------------|--------------------------------------------|
| 1     | Identify installation, operator, CN-coded goods and reporting period         | CASE and GOODS registers                   |
| 2     | Define production processes, source streams, boundaries and functional units | Monitoring Plan and System Boundary        |
| 3     | Capture direct, electricity and precursor data with approved evidence        | INPUTS, PRECURSORS and EVIDENCE            |
| 4     | Apply unit conversions, allocation and embedded-emissions formulae           | CALCULATION_TRACE and root hash            |
| 5     | Run evidence, methodology, reconciliation and integrity controls             | QUALITY_CONTROLS and MONITORING_PLAN       |
| 6     | Apply per-good 5% materiality reference for verifier planning                | GOODS materiality columns                  |
| 7     | Freeze, hash, sign and commit immutable package                              | Manifest, KMS signature and package hash   |
| 8     | Independent accredited verifier performs review and records opinion          | VERIFIER_SIGN_OFF — initially NOT_REVIEWED |

## System Boundary - Declared system boundary

|          |                                                                                                                                                    |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| BOUNDARY | Coke preparation, sinter plant, blast furnace, basic oxygen furnace, casting and finishing operations within the controlled installation boundary. |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------|

## System Boundary - Sector methodology

| Sector         |          |                                                                                                                                                                                         |                                                                                                                                                                               |
|----------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Iron and Steel | IN_FORCE | Coke oven, sinter or pellet preparation, blast furnace, basic oxygen furnace, direct reduction, electric arc furnace, casting and finishing processes within the installation boundary. | Mass balance and production ledger; Fuel and reducing-agent consumption; Precursor embedded emissions; Waste-gas and transferred-heat treatment; Allocation to CN-coded goods |

## Methodology - Decision register

| Topic                      |                                                                                      |                                                                                                                                                   |                                                                                                                                         |          |                                    |
|----------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------|
| PRECURSOR_SC_OPE           | No qualifying precursor line is declared for the fixture goods and controlled route. | The controlled bill of materials and production route contain no separate precursor goods requiring an additional precursor line in this fixture. | Regulation (EU) 2023/956 Annex IV and Commission Implementing Regulation (EU) 2025/2547; supported by the controlled bill of materials. | ACCEPTED | 11111111-1111-4111-8111-1111111111 |
| GOODS_EMISSIONS_ALLOCATION | Allocate installation emissions using documented product mass shares of 0.6 and 0.4. | The two goods are produced in the same controlled period and the reconciled product ledger provides complete mass shares summing to one.          | Commission Implementing Regulation (EU) 2025/2547 allocation and monitoring-plan requirements; controlled production ledger.            | ACCEPTED | 11111111-1111-4111-8111-1111111111 |

## Calculation Annex - Engine identity

# Technical Compilation and Calculation Annex

Consolidated CBAM evidence dossier and preparation statements

Report report\_aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa · Release 5

AUTOMATED PREPARATION CHECKS  
PASSED

CONFIDENTIAL - VERIFIER PREPARATION WORKSPACE

| Control                         |                                                                  |
|---------------------------------|------------------------------------------------------------------|
| Ruleset                         | EU-CBAM-DEFINITIVE-2026                                          |
| Engine version                  | 3.0.0                                                            |
| Calculation root hash           | 7e12ba9592177cc10e0e4541bb103905e0345303fa794949881b86ec2ab48f61 |
| Allocation share total          | 1                                                                |
| Allocation reconciliation delta | 0                                                                |

## Calculation Annex - Formula trace

| Formula                                    |     |                        |                                                                  |                                                                                                    |
|--------------------------------------------|-----|------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| CBAM_INDIRECT_EMISSIONS                    | 40  | tCO2e                  | 6a2647d365280b1a97c749006dfaa066e9374f7534b8c11cafe32b791c010350 | None                                                                                               |
| CBAM_PRECURSOR_EMISSIONS_SUM               | 0   | tCO2e                  | 5553fb3ff7a2d2d3f04af35bdc6f4c64f77ee3359321812bc30163f210d3c86  | None                                                                                               |
| CBAM_TOTAL_EMBEDDED_EMISSIONS              | 120 | tCO2e                  | cc192e8f6a78218e061c33a7c0a99f82059fba5ee1def6dd1735ccabc099b2d2 | None                                                                                               |
| CBAM_GOOD_EMISSIONS_ALLOCATION_1           | 1.2 | tCO2e/t                | 7aba824be81004f963dadb979458428808b42ab7d4f48dbfcae56a4f661fd5ba | None                                                                                               |
| CBAM_GOOD_EMISSIONS_ALLOCATION_2           | 1.2 | tCO2e/t                | 2d6d72ee83e4cd16157fe19d8f21971c0c6b8e36076c5bbbf42403b5a39f5a9b | None                                                                                               |
| CBAM_GOODS_ALLOCATION_RECONCILIATION       | 0   | tCO2e                  | 274077c9bd67745642e0aaf8506b146e80db2fca464cbcbf55485d95475c2fd0 | None                                                                                               |
| CBAM_AGGREGATE_SPECIFIC_EMBEDDED_EMISSIONS | 1.2 | tCO2e/t                | 61fff4226612cedc54b2fdf528abe1b84fd63436a2cef5e27dc6b2f41761f2d0 | The aggregate intensity is a portfolio diagnostic; reportable values are the per-good intensities. |
| CBAM_CARBON_PRICE_REDUCTION_RECORD_TOTAL   | 0   | certificate-equivalent | 13eee823710e59782162be2fd529ca7dd9bb3ac7b195d5d21486017266c848ea | None                                                                                               |

## Calculation Annex - Per-good reconciliation

| Good |          |                |    |     |    |     |      |
|------|----------|----------------|----|-----|----|-----|------|
| 1    | 72011011 | IRON_AND_STEEL | 60 | 0.6 | 72 | 1.2 | 0.06 |
| 2    | 72011019 | IRON_AND_STEEL | 40 | 0.4 | 48 | 1.2 | 0.06 |